

PIERRE, SD, USA

WMO: 726560

Lat: 44.381N

Lon: 100.286W

Elev: 1742

StdP: 13.79

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24025

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-6.8	-17.9	2.2	-10.9	-13.3	2.8	-6.2	35.0	19.2	32.1	21.3	8.1	310	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	96.8	70.8	92.7	70.4	88.9	69.3	75.6	87.8	73.7	85.8	71.9	84.1	13.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	127.5	81.7	70.1	118.0	80.2	68.0	109.9	78.2	40.4	87.7	38.5	85.6	36.7	84.3	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.0	26.6	23.8	-19.3	104.0	6.5	4.4	-24.0	107.2	-27.8	109.8	-31.5	112.2	-36.2	115.4		
			-19.6	79.2	6.4	2.2	-24.2	80.7	-28.0	82.0	-31.6	83.2	-36.3	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	18.4	22.0	33.7	46.2	57.5	68.1	75.3	73.0	64.1	49.0	34.4	22.2	(6)
(7)		DBStd	22.43	13.82	14.03	13.56	10.32	8.49	6.96	6.55	6.36	8.94	9.64	11.03	12.12	(7)
(8)		HDD50	3978	980	785	519	187	29	0	0	5	136	475	862	(8)	
(9)		HDD65	7380	1445	1205	969	568	260	46	6	11	122	504	917	1327	(9)
(10)		CDD50	2932	0	0	16	73	261	544	784	713	429	104	8	0	(10)
(11)		CDD65	855	0	0	0	3	27	139	325	258	96	7	0	0	(11)
(12)		CDH74	9316	0	0	14	90	379	1425	3521	2637	1133	115	2	0	(12)
(13)		CDH80	4129	0	0	2	24	108	523	1737	1220	478	37	0	0	(13)
(14)	Wind	WSAvg	10.8	10.8	10.8	11.7	12.3	11.6	10.2	9.5	9.7	10.6	11.2	10.7	10.8	(14)
(15)	Precipitation	PrecAvg	18.9	0.5	0.5	1.0	2.0	3.0	3.5	2.6	1.7	1.7	1.3	0.6	0.5	(15)
(16)		PrecMax	31.9	2.3	2.3	3.7	6.4	7.2	7.3	6.4	5.2	6.9	5.4	2.6	1.2	(16)
(17)		PrecMin	7.1	0.0	0.0	0.0	0.2	0.6	0.5	0.2	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.8	0.4	0.5	0.8	1.5	1.6	1.9	1.7	1.2	1.5	1.2	0.6	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	60.8	75.1	82.4	88.6	96.2	105.4	100.4	98.2	85.1	69.5	54.3	(19)
(20)		MCWB	42.5	45.8	53.4	58.9	67.0	72.1	71.4	70.2	68.2	61.6	52.1	42.3	(20)	
(21)		2%	DB	45.0	51.8	65.9	75.0	82.6	89.8	98.0	95.2	90.6	75.7	61.4	46.4	(21)
(22)		MCWB	37.8	41.4	50.0	56.2	63.9	70.5	71.4	70.9	66.9	58.1	47.6	38.6	(22)	
(23)		5%	DB	40.3	45.2	59.0	69.1	78.2	85.8	93.4	91.0	85.3	69.8	55.2	41.4	(23)
(24)		MCWB	35.2	38.6	46.9	52.8	61.9	68.5	71.3	70.0	65.2	55.1	44.8	35.7	(24)	
(25)		10%	DB	36.8	39.9	53.0	63.6	73.3	82.2	89.3	86.9	80.7	64.0	49.8	37.5	(25)
(26)		MCWB	33.1	35.2	44.3	49.5	59.3	67.2	70.5	68.5	63.7	52.6	41.8	33.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	47.4	55.5	61.2	70.2	77.0	78.5	77.2	72.5	64.6	53.1	43.8	(27)
(28)		MCDWB	53.0	58.2	68.5	77.8	83.1	88.3	90.5	89.6	85.0	78.2	66.3	52.5	(28)	
(29)		2%	WB	38.5	42.2	51.4	57.0	66.8	73.7	76.4	74.6	70.1	60.2	48.9	39.0	(29)
(30)		MCDWB	44.4	50.4	63.6	72.9	79.2	85.5	88.6	86.6	84.4	72.4	60.2	45.9	(30)	
(31)		5%	WB	35.4	38.7	47.9	53.8	63.6	71.3	74.4	72.6	68.2	56.6	45.3	35.9	(31)
(32)		MCDWB	39.7	44.8	58.6	66.4	74.8	82.4	86.9	84.3	81.3	67.6	54.5	41.0	(32)	
(33)		10%	WB	33.1	35.4	43.9	51.0	60.6	68.9	72.5	70.7	65.8	53.4	42.1	33.4	(33)
(34)		MCDWB	36.7	39.7	52.7	61.9	71.1	79.4	85.2	82.7	77.7	62.7	49.4	37.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	20.3	21.6	22.3	24.0	23.4	23.3	25.7	25.4	25.8	22.5	20.9	18.5	(35)	
(36)		5% DB	MCDBR	24.0	27.7	33.4	33.8	30.3	28.3	30.6	30.8	32.8	32.1	29.4	24.2	(36)
(37)		MCWBR	17.9	18.7	19.0	17.7	15.9	13.1	11.2	12.2	13.6	16.6	17.7	17.3	(37)	
(38)		5% WB	MCDBR	23.6	26.9	32.3	31.4	27.1	24.9	26.6	26.7	28.1	28.2	28.4	23.4	(38)
(39)		MCWBR	18.2	18.6	19.0	17.3	15.5	13.6	12.3	12.9	13.1	16.0	17.7	17.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.261	0.270	0.293	0.340	0.362	0.373	0.389	0.386	0.349	0.306	0.273	0.259	(40)	
(41)		taud	2.514	2.510	2.519	2.380	2.394	2.394	2.378	2.368	2.450	2.541	2.542	2.510	(41)	
(42)		Ebn at Noon	281	298	302	293	288	284	278	275	278	279	273	270	(42)	
(43)		Edh at Noon	21	25	29	36	37	37	37	36	31	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	595	901	1235	1565	1884	2131	2187	1826	1457	957	640	482	(44)	
(45)		RadStd	50	77	95	117	179	135	82	144	96	118	67	30	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (11 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47)

Nomenclature: See separate page